

Southwest
Tech

Python Programming Foundations

Week 5 - Files, Errors, & Data Persistence
Day 3: "Comparing Data Representations & Application Structure"

1

Same Information,
Different Forms

CSV shape helps with rows and columns.

JSON shape helps with labels and nested information.

Different shapes make different tasks easier.

2

Southwest
Tech

One Study Task, Four Ways

Same information, different representations

Plain note

Task: Read chapter
Topic: Files
Minutes: 30

CSV row

Task	Topic	Minutes
Read chapter	Files	30

JSON object

```
{
  "Task": "Read chapter",
  "Topic": "Files",
  "Minutes": 30
}
```

Table row

Task	Topic	Minutes
Read chapter	Files	30

Different formats, same meaning.
Knowing how data can be represented helps you read it, work with it, and explain it.

Same Information, Different Forms

3

Review: Structure Data Has Shape

- ▶ CSV shape helps with rows and columns.
- ▶ JSON shape helps with labels and nested information.
- ▶ Different shapes make different tasks easier.

4

Southwest
Tech

Structured Data Has Shape

CSV

rows and columns

Task	Topic	Minutes
Read chapter	Files	30

good for similar records

JSON

labeled grouped values

```
{
  "Task": "Read chapter",
  "Topic": "Files",
  "Minutes": 30
}
```

good for labeled structure

Different shapes make different tasks easier.

Shape of Data (Structured)

5

Southwest
Tech

What Counts as Success Today

- recognize at least two representations
- compare what each makes easier
- compare what each makes harder
- connect the representation to a larger application need
- avoid treating advanced structures as automatically better

6

Data Representation Success Pattern

Compare forms using the same information.

1. recognize forms
How can this information be represented?
CSV • JSON • Table • Note • Others

2. what gets easier?
What tasks become simpler with this representation?
• Read
• Filter
• Summarize
• Update

3. what gets harder?
What tasks become more difficult with this representation?
• Look up by key
• Add new fields
• Handle variation

4. larger apps need
What will the larger application need to do?
• Speed
• Flexibility
• Integration
• Maintenance

Task: Read chapter
Topic: Files
Minutes: 30

Representation choice depends on use.

Today's Success

7

What We Will Use Today

- ▶ plain text
- ▶ CSV
- ▶ JSON
- ▶ list/dictionary structures
- ▶ table-like or model-like previews

We will ask what each form makes easier or harder.

8

Toolbox

Some information when in plain simple forms.

plain text
Task: Read chapter
Topic: Files
Minutes: 30

CSV
Task: Read chapter
Topic: Files
Minutes: 30

JSON
Task: Read chapter
Topic: Files
Minutes: 30

list/dictionary
Task: Read chapter
Topic: Files
Minutes: 30

table-like preview
Field Value
Task: Read chapter
Topic: Files
Minutes: 30

What becomes easier?
Which tasks are simpler or more natural in some forms?

What becomes harder?
Which tasks are trickier or require more work in some forms?

REPRESENTATION CHOICE DEPENDS ON USE.

9

What We Will "Skip" For Now

- ▶ Building a database
- ▶ Writing SQL
- ▶ Implementing an ORM
- ▶ Designing a production data model

Today is recognition and comparison.

10

Scope Boundary

Focus on what helps us most right now.

1 Lane
Today: recognize and compare
plain text • CSV • JSON • list/dictionary • table-like preview
Goal: Understand the shapes and how they compare.

2 Lane
Later: build databases or ORMs
Designs schemas, relationships, and ORMs when we start building larger applications.
Saved for later: Important and useful when we need it.

We focus on the next right step without losing sight of what comes next.

Parked For Later

11

Data Forms

12

Plain Text Is Human Friendly



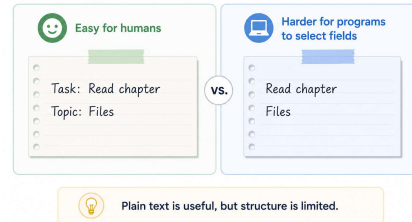
Plain text is easy for humans to read.



It becomes harder when the program needs to reliably select fields, filter records, or compare values.

13

Plain Text



“Human” Friendly:
Plain Text

14

CSV Is Table Friendly

CSV works well when data fits rows and columns.

It is useful for lists of similar records, but it can get awkward when data needs deeper nested structure.

CSV

Comma-Separated Values

Task	Topic	Minutes
Read chapter	Files	30
Review notes	Data shapes	20
Practice examples	Algorithms	45



Good for similar records

Works well when each row has the same fields.



Awkward for deeply nested data

Hard to represent complex or hierarchical relationships.

“Table” Friendly:
CSV

16

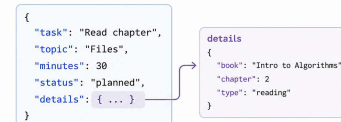
JSON Is Structure Friendly

JSON works well when data needs labels, groups, or nested pieces.

It can be very useful, but it can also become hard to read if the nesting gets too deep.

JSON

JavaScript Object Notation



Good for labeled groups

Names make the data easy to understand and select.



Can get hard to read if nesting grows

More levels add detail but can reduce clarity.

“Structure” Friendly:
JSON

18

17

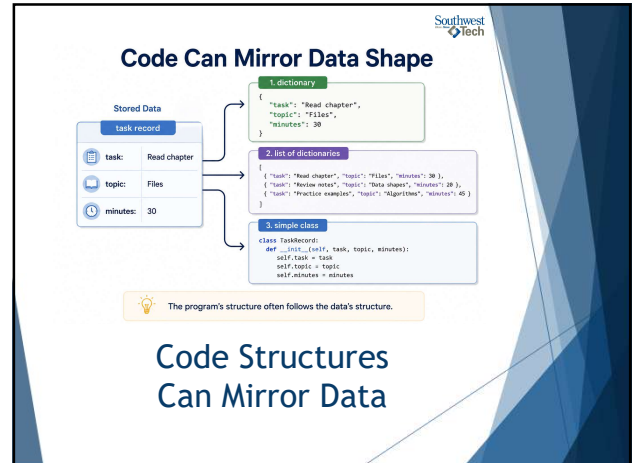


Code Structures Can Mirror Data

Lists, dictionaries, and simple classes can mirror the shape of stored data.

The program's structure often follows the data's structure.

19



Code Can Mirror Data Shape

Stored Data

```
task record
task: Read chapter
topic: Files
minutes: 30
```

1. dictionary

```
{
  "task": "Read chapter",
  "topic": "Files",
  "minutes": 30
}
```

2. list of dictionaries

```
[
  { "task": "Read chapter", "topic": "Files", "minutes": 30 },
  { "task": "Review notes", "topic": "Data shapes", "minutes": 20 },
  { "task": "Practice examples", "topic": "Algorithms", "minutes": 40 }
]
```

3. simple class

```
class TaskRecord:
    def __init__(self, task, topic, minutes):
        self.task = task
        self.topic = topic
        self.minutes = minutes
```

The program's structure often follows the data's structure.

Code Structures Can Mirror Data

20

Better Depends On Use

? Ask:

- What needs to be read?
- What needs to be searched?
- What needs to be updated?
- What needs to be explained?

The answers guide the representation choice.

21

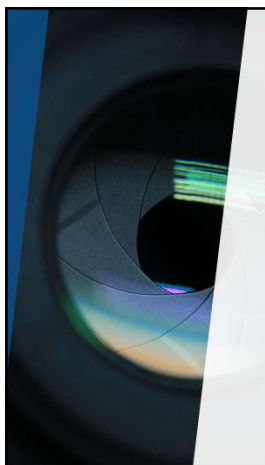
Data Representation Decision

Answer these questions to guide your choice.

- What needs to be read?
- What needs to be searched?
- What needs to be updated?
- What needs to be explained?

Better Depends On Use

22



Demo

"Basic Type Awareness"

23

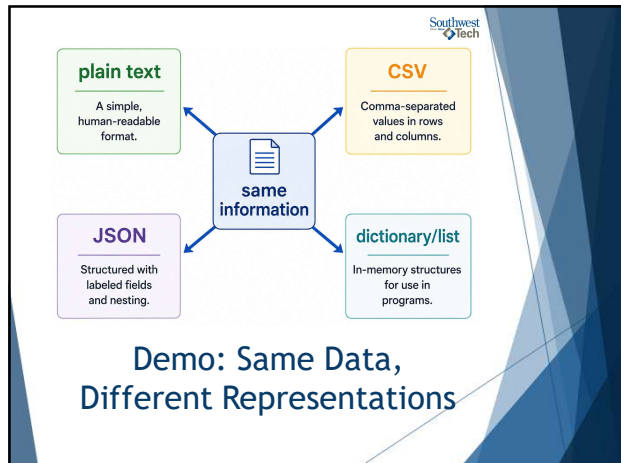


What to Watch For

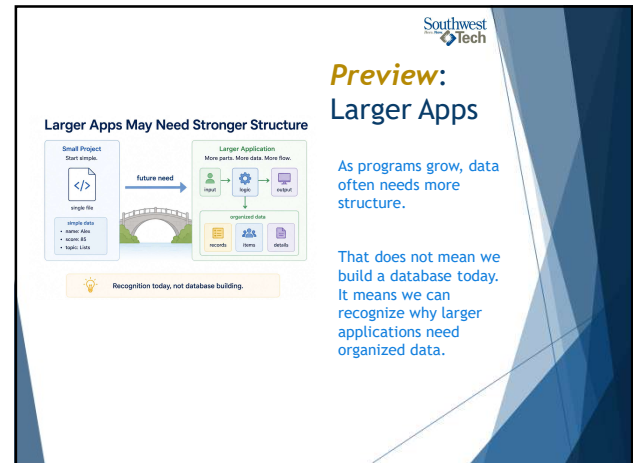
Watch what changes when the same information appears as:

- plain text
- CSV-style data
- JSON-style data
- dictionary/list structure

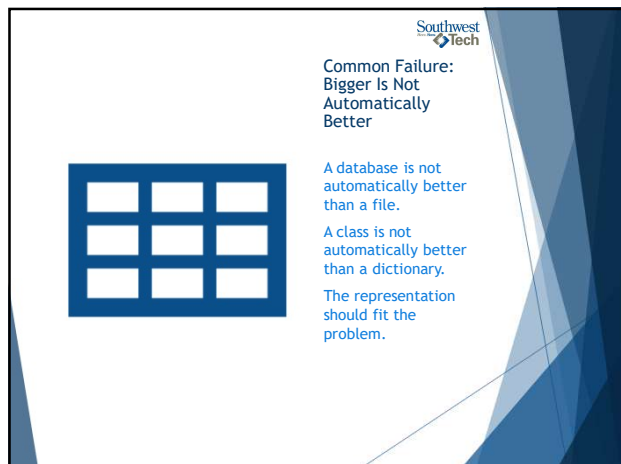
24



25



26



27



28



29

A10: Compare and Explain

Compare two representations of the same information and reflect on the differences.

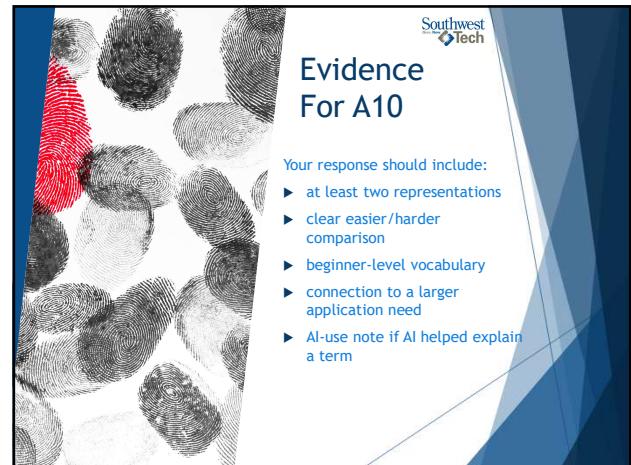
- Representation 1
- Representation 2
- What became easier?
- What became harder?
- Where might a larger app need more structure?

Assignment 10

30

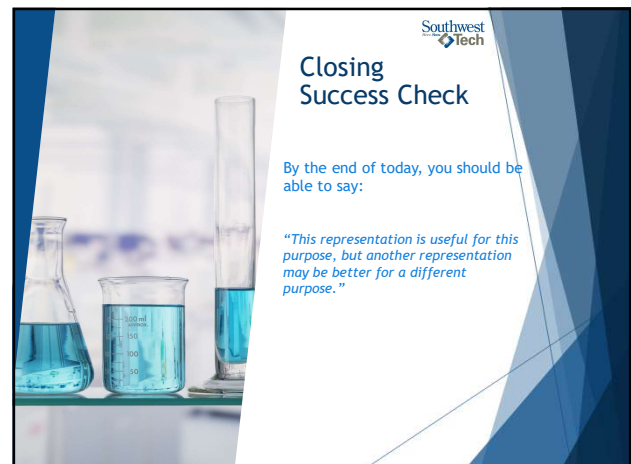


31



32

33



34



35